

Daizhi Liao

Undergraduate researcher · Reinforcement Learning & Embodied AI · PhD applicant Fall 2027

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EDUCATION

Guangzhou University — B.Eng., Network Engineering

Expected 2027

Third-year undergraduate. Focus: reinforcement learning, embodied AI, and the empirical auditing of deployed AI systems.

RESEARCH & PROJECTS

TransitTruth — Behavioral-Fingerprint Audit of AI API Relays

2026

Solo design & implementation. Open-source tool that verifies whether an AI API relay actually serves the advertised model, by probing statistical behavior (random-number, token, latency distributions) against baselines with chi-square / KS tests and Bayesian updating. Built a zero-install, fully client-side web demo and a 20-page technical report validating same-family discrimination: gpt-4o vs gpt-4o-mini, TVD up to 0.90, 91.3% in-distribution resubstitution posterior (not held-out).

Code & demo: github.com/dafahaha/transit-truth · live demo

embodied-offpolicy-study — From-Scratch SAC Ablation on Hopper-v4

2026

Soft Actor-Critic implemented from scratch in PyTorch (no Stable-Baselines3), run as a pre-registered, multi-seed ablation (CPU only, 60k env steps/run, 3 seeds per arm): does scaling the rewards entering the critic's Bellman backup change sample efficiency? Honest result — the reward-scaled arm trends higher (asymptotic return ~1087 vs ~363) but the 95% CIs overlap heavily at n=3, so this is a directional signal, not a confirmed effect or SOTA. Raw CSVs are committed and the learning curve is reproduced by script.

Code: github.com/dafahaha/embodied-offpolicy-study

rl-deploy-bench — Cross-Platform RL Deployment & Benchmarking

v1.1.0

Config-driven CLI: export Stable-Baselines3 policies to ONNX/TorchScript, build TensorRT engines with FP16/INT8 quantization, and benchmark latency, throughput, and accuracy across x86 GPU, NVIDIA Jetson, and CPU, with auto-generated HTML reports. A reproducible harness for the accuracy-latency tradeoff of quantized RL policies.

OPEN SOURCE — MERGED CONTRIBUTIONS

- **microsoft/TextWorld #377** (merged) — Fixed env. step crash on multi-command input; non-greedy score parsing plus regression test.
- **shmuma/ptan #57** (merged) — Fixed Gymnasium API compatibility in experience sources.
- **Algorineko/AgenticArXiv-RL #72** (merged) — Added MIT License, CONTRIBUTING guide, issue templates, and CI.
- Additional review-phase bug fixes upstream in vLLM, pandas, google/brax, Farama-Foundation/PettingZoo, and Hugging Face datasets & diffusers.

TECHNICAL SKILLS

RL & learning: PyTorch, Stable-Baselines3, Gymnasium, MuJoCo | **Deployment:** ONNX, ONNX Runtime, TensorRT, TorchScript, INT8/FP16 quantization | **Systems:** Python, C++, Linux, Docker, Git, CI/CD, CUDA